



Technical Service Bulletin: TSB220210

Released Date: 05-Jun-2025

Improved Injector Upper Plunger Assembly

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Warranty Statement

The information in this document has no effect on present warranty coverage or repair practices, nor does it authorize TRP or Campaign actions.

Contents

Product Affected

- QSK19 CM500
- QSK23 CM500
- QSK23 CM2250 K109
- QSK45 CM500
- QSK60 CM500
- QSK60 CM2250 K112
- QSK78 CM500
- QSK78 CM2250 K104

Issue Summary

Symptom:

- Power cylinder malfunction
- Increased exhaust gas temperatures: if using PrevenTech®, a warning will be issued by the PrevenTech® team to check the affected cylinder.

Root Cause:

- Upper injector plunger couplings have been manufactured with oversized press diameters. This resulted in an inadequate retention force which could **not** withstand the separating forces generated by the operation of the injector.

Verification

- Visually the injector appears undamaged.
- The upper plunger assembly can **only** be verified after disassembling the injector. A good injector will have the coupling and plunger removed as a single item, for a malfunctioning upper plunger assembly the coupling and plunger will be separated. A comparison of the upper plunger assembly is shown in Figure 1.



Figure 1, 1: Non-malfunctioning Upper Plunger Assembly. 2: Malfunctioning Upper Plunger Assembly.

Resolution

- Injector manufacturing issue has been resolved at the factory.

Service Instructions

- All parts on hand **must** be examined and serial numbers on injectors checked to verify if they align to the clean date. See Part Identification section below.
- All suspect injectors **must** be quarantined and returned to Cummins Inc. Fuel Systems. See Part Inventory Action section below.

Part Identification

Table 1, Impacted Injector Part Numbers and Serial Number Ranges			
Injector Part Number	Engine Model	Suspect Injector Serial Number Range	Figure Example
4326639	QSK78	Serial number between 1T20183XXXX to 1T22132XXXX	2
4326640			
4928345	QSK19		
4928346			
4928347			
4928348			
4928349			
4088431			
4954892			
4088432			
4902827			
4902828			
4954921			
4954990			
4326779	QSK45 QSK60		
4326780			
4326781			
4326782			
4326783			
4326784			

Table 1, Impacted Injector Part Numbers and Serial Number Ranges			
Injector Part Number	Engine Model	Suspect Injector Serial Number Range	Figure Example
4326639PX	QSK78	Serial number between 070120XXX to 051622XXX	4
4326640PX			
4928345RX	QSK19		
4928346RX			
4928347RX			
4928348RX			
4928349RX			
4088431PX	QSK23		
4954892PX			
4902827PX			
4902828RX			
4326779PX	QSK45 QSK60		
4326780PX			
4326781PX			
4326782PX			
4326783PX			
4326784PX			

- Any injectors with a serial number between 1T20183XXXX to 1T22132XXXX **must** be quarantined.
- Additionally, the overhaul kits listed in Table 2 contain impacted injector part numbers.

Table 2, Impacted Overhaul Kits		
Overhaul Kit Part Number	Engine Model	Injector Part Number
5579434RX	QSK23	4954892PX
5579385RX		4088431PX
5579386RX		4954892PX
5579433RX		4954892PX
5579387RX		4902827PX



Figure 2, New Injector Serial Number Labelling on Top Stop.

- In addition, the box in which new injectors are shipped has a label attached stating the injectors use 'Certified upper plunger couplings' as shown in Figure 3.



Figure 3, Label Attached to Injector Shipping Box.

- Any injectors with a serial number between 070120XXX to 051622XXX **must** be quarantined.

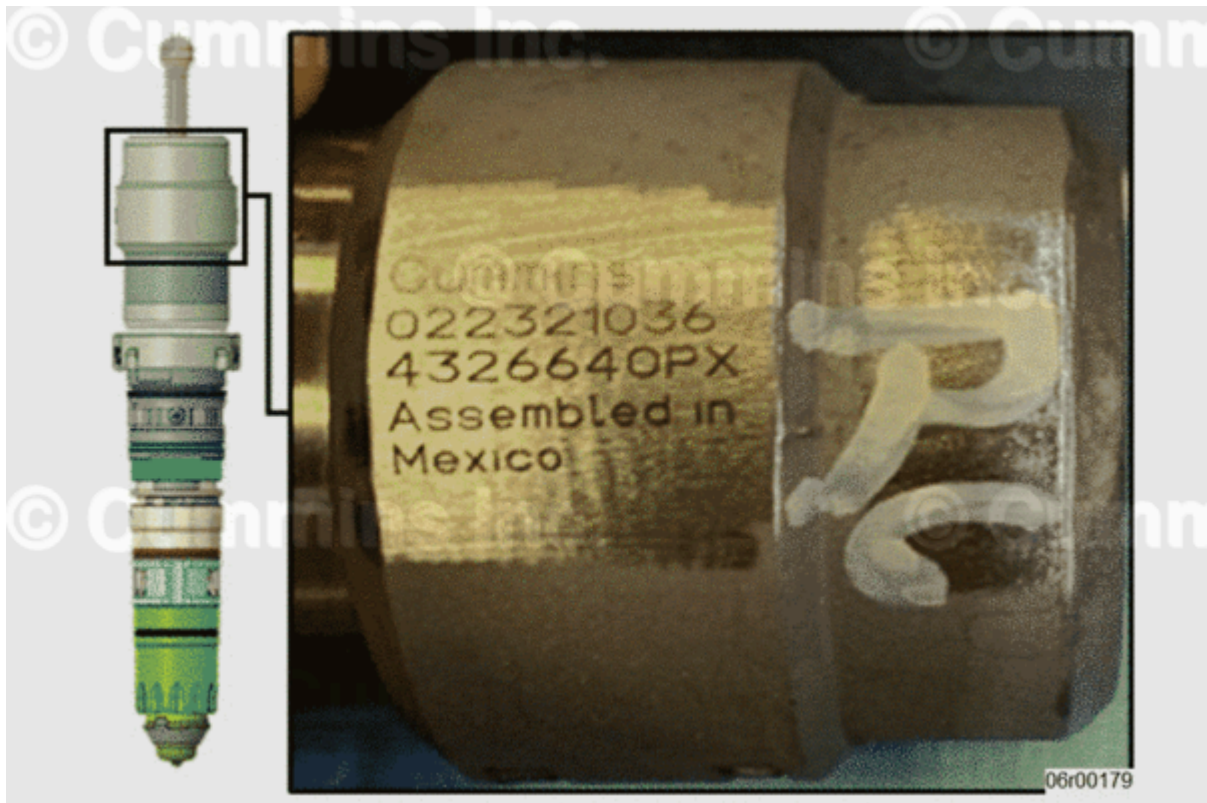


Figure 4, Recon® Injector Serial Number Labelling on Top Stop.

- In addition to the serial number, a green dot has also been applied to the top stop on the injector, the individual injector box and the final shipping box as shown in Figure 5.

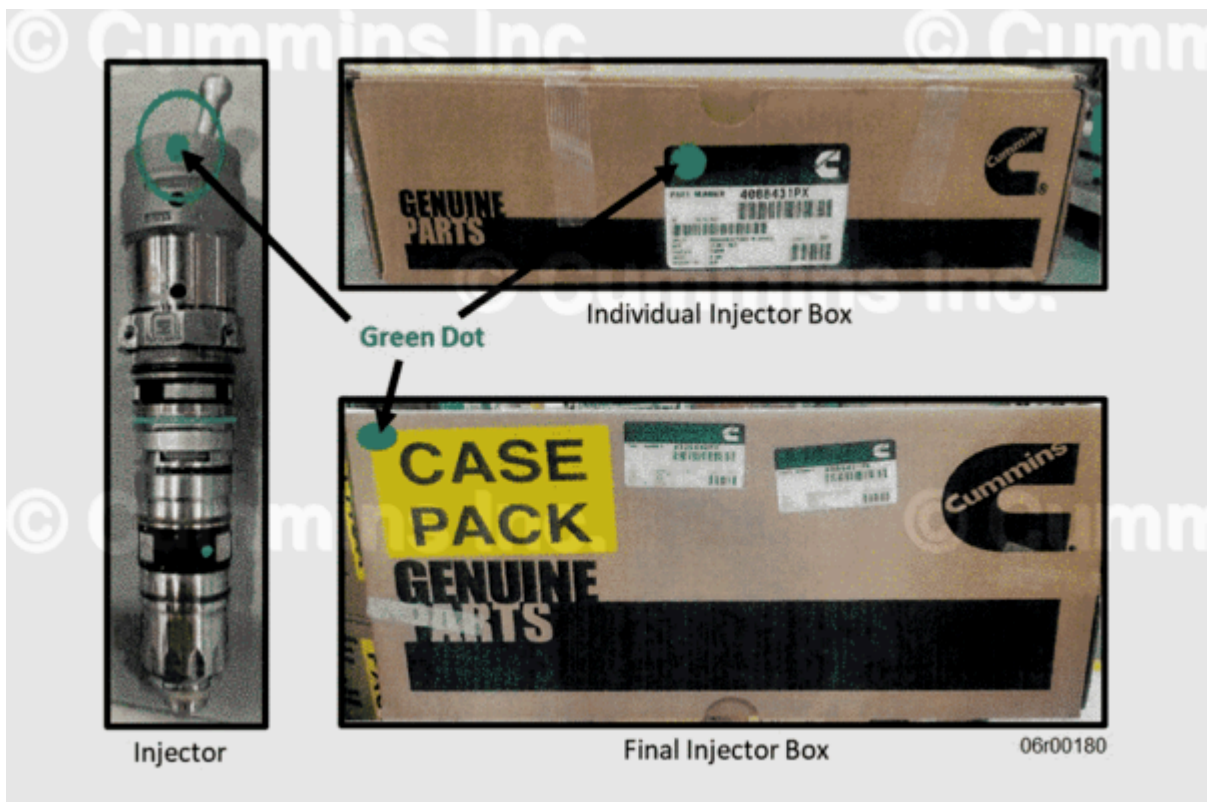


Figure 5, Certified Recon® Injector Identification.

- In stock parts are affected.
- For QSK23 injectors **only**, see Parts Alert 2024-155 for details.
- For QSK19, QSK45, QSK60, QSK78 injectors, use existing inventory before using new parts.

Document History

Date	Details
2022-10-11	Module Created
2022-11-30	Updated Product Affected and Table 1 serial number range.
2024-4-16	Updated Product Affected and Table 1.
2024-12-10	Updated tsb to include Table 2 and updated Parts Inventory Action section
2025-6-3	Updated Part Inventory Action

Last Modified: 05-Jun-2025